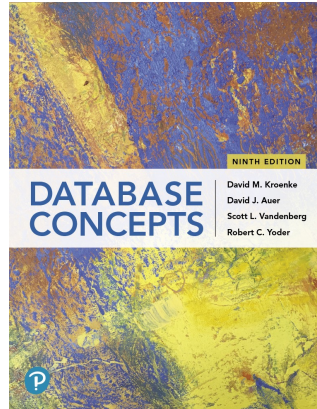


Database Concepts

Ninth Edition



Chapter 1

Getting Started

Pearson

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1

Learning Objectives (1 of 2)

- Understand the importance of databases in Internet Web applications and mobile apps
- Understand the nature and characteristics of databases
- Understand the potential problems with lists
- Understand the reasons for using a database
- Understand how using related tables helps you avoid the problems of using lists
- Know the components of a database system
- Learn the elements of a database
- Learn the purpose of a database management system (DBMS)
- Understand the functions of a database application

Pearson

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2

Learning Objectives (2 of 2)

- Introduce Web database applications
- Introduce data warehouses and business intelligence (BI) systems
- Introduce Big Data and cloud computing



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3

The Importance of Databases in the Internet and Mobile App World

Understand the importance of databases in Internet Web applications and mobile apps

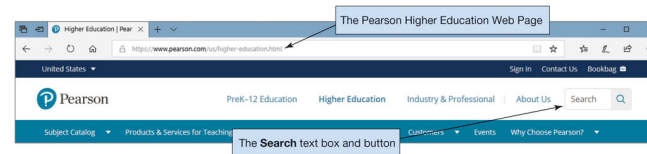
- The Internet was created as the ARPANET in 1969 and grew to connect LANs.
- It became known and used when the World Wide Web (Web) became accessible in 1993.
- In the early 2000s, Web 2.0 sites started to appear allowing users to add content.
- The Internet of Things (IoT) is the latest development which allows all types of devices to connect with each other.
- All of these items depend upon databases.



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Figure 1.1(a) Searching a Database in a Web Browser



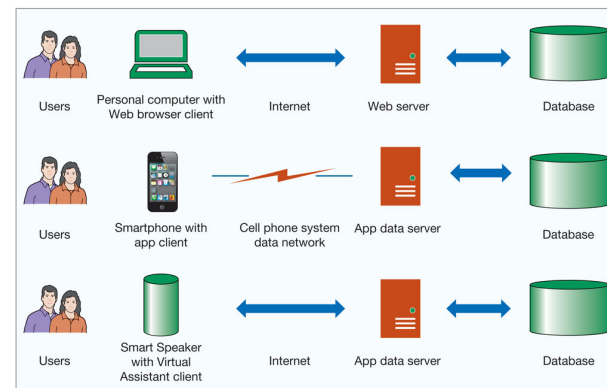
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Figure 1.2 The Internet and Mobile Device World



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Why Use a Database?

Understand the nature and characteristics of databases

- The reason databases are used is to keep track of things.
- Databases store more complicated information than simple lists like a spreadsheet.

Figure 1.7 The Student with Advisor and Department List

Understand the potential problems with lists

- A major problem with using lists include:
 - **Modification problems:** Redundancy and multiple themes can create modification problems such as deleting, updating, and inserting records as seen in the figure below.

If Adviser Baker is changed to Taling, we need to change AdviserEmail as well. If changed to Valdez, we need to change AdviserEmail, Department, and AdminLastname.

	A	B	C	D	E	F	G	H
	StudentLastname	StudentFirstname	StudentEmail	AdviserLastname	AdviserEmail	Department	AdviserLastname	
1	145	Andrew	Andrew	Andrew	Andrew	Accounting	Smith	
2	14500	Fischer	Douglas	Douglas	Douglas	Accounting	Smith	
3	14500	Thompson	James	James	James	Accounting	Smith	
4	14500	Thompson	James	James	James	Accounting	Smith	
5	14500	Thompson	James	James	James	Accounting	Smith	
6	14500	Thompson	James	James	James	Accounting	Smith	
7	14500	Thompson	James	James	James	Accounting	Smith	
8	14500	Thompson	James	James	James	Accounting	Smith	
9	14500	Thompson	James	James	James	Accounting	Smith	
10	14500	Thompson	James	James	James	Accounting	Smith	
11	14500	Thompson	James	James	James	Accounting	Smith	
12	14500	Thompson	James	James	James	Accounting	Smith	
13	14500	Thompson	James	James	James	Accounting	Smith	
14	14500	Thompson	James	James	James	Accounting	Smith	
15	14500	Thompson	James	James	James	Accounting	Smith	
16	14500	Thompson	James	James	James	Accounting	Smith	
17	14500	Thompson	James	James	James	Accounting	Smith	
18	14500	Thompson	James	James	James	Accounting	Smith	
19	14500	Thompson	James	James	James	Accounting	Smith	
20	14500	Thompson	James	James	James	Accounting	Smith	
21	14500	Thompson	James	James	James	Accounting	Smith	
22	14500	Thompson	James	James	James	Accounting	Smith	
23	14500	Thompson	James	James	James	Accounting	Smith	
24	14500	Thompson	James	James	James	Accounting	Smith	
25	14500	Thompson	James	James	James	Accounting	Smith	
26	14500	Thompson	James	James	James	Accounting	Smith	
27	14500	Thompson	James	James	James	Accounting	Smith	
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30	14500	Thompson	James	James	James	Accounting	Smith	
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32	14500	Thompson	James	James	James	Accounting	Smith	
33	14500	Thompson	James	James	James	Accounting	Smith	
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35	14500	Thompson	James	James	James	Accounting	Smith	
36	14500	Thompson	James	James	James	Accounting	Smith	
37	14500	Thompson	James	James	James	Accounting	Smith	
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84	14500	Thompson	James	James	James	Accounting	Smith	
85	14500	Thompson	James	James	James	Accounting	Smith	
86	14500	Thompson	James	James	James	Accounting	Smith	
87	14500	Thompson	James	James	James	Accounting	Smith	
88	14500	Thompson	James	James	James	Accounting	Smith	
89	14500	Thompson	James	James	James	Accounting	Smith	
90	14500	Thompson	James	James	James	Accounting	Smith	
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92	14500	Thompson	James	James	James	Accounting	Smith	
93	14500	Thompson	James	James	James	Accounting	Smith	
94	14500	Thompson	James	James	James	Accounting	Smith	
95	14500	Thompson	James	James	James	Accounting	Smith	
96	14500	Thompson	James	James	James	Accounting	Smith	
97	14500	Thompson	James	James	James	Accounting	Smith	
98	14500	Thompson	James	James	James	Accounting	Smith	
99	14500	Thompson	James	James	James	Accounting	Smith	
100	14500	Thompson	James	James	James	Accounting	Smith	

Deleted row—Student and Adviser data lost

Inserted row—both Student and Adviser data missing

Excel 2019, Windows 10, Microsoft Corporation.

Using Relational Database Tables

Understand how using related tables helps you avoid the problems of using lists.

- **Relational model** is a methodology used as a solution for database design.
- A **relational database** contains a collection of separate tables.
- A **table** holds data about only one theme.
- Each **column**, also known as fields, in a table stores a characteristic common to all rows in a table. An example is StudentNumber.
- A **row** in a table, also known as a record, has data about an occurrence. An example would be all the information on one student.

Figure 1.8 The Advisor and Student Tables

STUDENT data linked to ADVISER data via AdviserLastName

The image shows two Microsoft Access tables. The top table is named 'ADVISER' and has columns: AdviserLastName, AdviserFirstName, and AdviserEmail. The bottom table is named 'STUDENT' and has columns: StudentLastName, StudentFirstName, StudentEmail, Phone, Residence, and StudentNumber. Red arrows indicate a relationship between the 'AdviserLastName' field in the 'ADVISER' table and the 'StudentLastName' field in the 'STUDENT' table. The 'STUDENT' table also has a 'StudentNumber' field which is highlighted in blue.

AdviserLastName	AdviserFirstName	AdviserEmail
Baker	Linda	Linda.Baker@ourcampus.edu
Green	George	George.Green@ourcampus.edu
Tang	Susan	Susan.Tang@ourcampus.edu
Tan	Ken	Ken.Tan@ourcampus.edu
Valdez	Richard	Richard.Valdez@ourcampus.edu

StudentLastName	StudentFirstName	StudentEmail	Phone	Residence	StudentNumber
Andrews	Matthew	Matthew.Andrews@ourcampus.edu	301-555-2266	123 12th St apt 21	10001
Fisher	Douglas	Douglas.Fisher@ourcampus.edu	301-555-2257	McKinley Room 109	10002
Hwang	Terry	Terry.Hwang@ourcampus.edu	301-555-2229	McKinley Room 208	10003
Thompson	James	James.Thompson@ourcampus.edu	301-555-2265	365 17th St apt 41	10004
Brisbon	Lisa	Lisa.Brisbon@ourcampus.edu	301-555-2281	Dorsett Room 201	10005
Lai	Tzu	Tzu.Lai@ourcampus.edu	301-555-2281	McKinley Room 115	10006
Morris	Chop	Chop.Morris@ourcampus.edu	301-555-2260	214 10th St apt 12	10007

Figure 1.9 Modifying the Advisor and Student Tables

Changed data—data remains consistent

Inserted data—no STUDENT data required

Deleted data—no ADVISER data lost

ADVISER

AdviserLastName	AdviserFirstName	AdviserEmail
Baker	Linda	Linda.Baker@ourcampus.edu
Green	George	George.Green@ourcampus.edu
Tang	Sue	Sue.Tang@ourcampus.edu
Tran	Ken	Ken.Tran@ourcampus.edu
Valdez	Richard	Richard.Valdez@ourcampus.edu
Travis	Bill	Bill.Travis@ourcampus.edu

STUDENT

SID	StudentLastName	StudentFirstName	StudentEmail	Phone	Residence	AdviserLastName
10023	Andrews	Matthew	Matthew.Andrews@ourcampus.edu	301-555-2225	123 15th St Apt 21	Baker
10060	Fischer	Douglas	Douglas.Fischer@ourcampus.edu	301-555-2227	McKinley Room 109	Baker
10088	Huang	Terry	Terry.Huang@ourcampus.edu	301-555-2229	McKinley Room 208	Tang
10132	Thompson	James	James.Thompson@ourcampus.edu	301-555-2245	345 17th St Apt 43	Tang
10154	Brisbon	Lisa	Lisa.Brisbon@ourcampus.edu	301-555-2241	Dorsett Room 201	Valdez
10167	Lai	Tzu	Tzu.Lai@ourcampus.edu	301-555-2231	McKinley Room 115	Valdez
10212	Marino	Chip	Chip.Marino@ourcampus.edu	301-555-2241	234 18th St Apt 32	Tran

Access 2019, Windows 10, Microsoft Corporation.



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Figure 1.10 The Department, Advisor, and Student Tables

Can insert DEPARTMENT data as needed—no ADVISER or STUDENT data required

Can change STUDENT Adviser name as needed—new value is linked to its own data

Can delete STUDENT data as needed—no DEPARTMENT or ADVISER data lost

DEPARTMENT

DepartmentName	DepartmentPhone	AdminLastName	AdminFirstName	AdminEmail
Biology	301-555-1001	Beth	Shawn	Shawn.Beth@ourcampus.edu
Chemistry	301-555-1002	Kelly	Chris	Chris.Kelly@ourcampus.edu
Information Systems	301-555-1003	Chapin	Robin	Robin.Chapin@ourcampus.edu
Accounting	301-555-1004	Regan	Aaron	Aaron.Regan@ourcampus.edu

ADVISER

AdviserLastName	AdviserFirstName	AdviserEmail	Department
Baker	Linda	Linda.Baker@ourcampus.edu	Accounting
Green	George	George.Green@ourcampus.edu	Biology
Tang	Sue	Sue.Tang@ourcampus.edu	Accounting
Tran	Ken	Ken.Tran@ourcampus.edu	Information Systems
Valdez	Richard	Richard.Valdez@ourcampus.edu	Chemistry
Travis	Bill	Bill.Travis@ourcampus.edu	Information Systems

STUDENT

SID	StudentLastName	StudentFirstName	StudentEmail	Phone	Residence	AdviserLastName
10023	Andrews	Matthew	Matthew.Andrews@ourcampus.edu	301-555-2225	123 15th St Apt 21	Baker
10060	Fischer	Douglas	Douglas.Fischer@ourcampus.edu	301-555-2227	McKinley Room 109	Baker
10088	Huang	Terry	Terry.Huang@ourcampus.edu	301-555-2229	McKinley Room 208	Tang
10132	Thompson	James	James.Thompson@ourcampus.edu	301-555-2245	345 17th St Apt 43	Tang
10154	Brisbon	Lisa	Lisa.Brisbon@ourcampus.edu	301-555-2241	Dorsett Room 201	Valdez
10167	Lai	Tzu	Tzu.Lai@ourcampus.edu	301-555-2231	McKinley Room 115	Valdez
10212	Marino	Chip	Chip.Marino@ourcampus.edu	301-555-2241	234 18th St Apt 32	Tran

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Figure 1.11 The Art Course List with Modification Problems

How to enter the fee for a new course?

Consequences of changing this date?

Consequences of deleting this row?

	A	B	C	D	E	F	G
1	CustomerLastName	CustomerFirstName	Phone	CourseDate	AmountPaid	Course	Fee
2	Johnson	Ariel	206-567-1234	10/1/2019	\$250.00	Adv Pastels	\$500.00
3	Green	Robin	425-678-8765	9/15/2019	\$350.00	Bag Oils	\$350.00
4	Jackson	Charles	360-789-3456	10/1/2019	\$500.00	Adv Pastels	\$500.00
5	Johnson	Ariel	206-567-1234	3/15/2019	\$350.00	Int Pastels	\$350.00
6	Pearson	Jeffery	206-567-2345	10/1/2019	\$500.00	Adv Pastels	\$500.00
7	Sears	Miguel	360-789-4567	9/15/2019	\$350.00	Bag Oils	\$350.00
8	Kyle	Leah	425-678-7654	11/15/2019	\$250.00	Adv Pastels	\$500.00
9	Myers	Lynda	360-789-5678	10/15/2019	\$0.00	Bag Oils	\$350.00

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Figure 1.12 The Art Course Database Tables

Can change COURSE CourseDate without problems

Can insert new COURSE data as needed

Can delete ENROLLMENT rows as needed – no adverse consequences

CUSTOMER			
CustomerNumber	CustomerLastName	CustomerFirstName	Phone
1	Johnson	Ariel	206-567-1234
2	Green	Robin	425-678-8765
3	Jackson	Charles	360-789-3456
4	Pearson	Jeffery	206-567-2345
5	Sears	Miguel	360-789-4567
6	Kyle	Leah	425-678-7654
7	Myers	Lynda	360-789-5678

COURSE			
CourseNumber	Course	CourseDate	Fee
1	Adv Pastels	10/1/2019	\$500.00
2	Bag Oils	9/15/2019	\$350.00
3	Int Pastels	3/15/2019	\$350.00
4	Bag Oils	10/15/2019	\$350.00
5	Adv Pastels	11/15/2019	\$500.00
(New)			\$0.00

ENROLLMENT			
CustomerNumber	CourseNumber	AmountPaid	
1	1	\$250.00	
2	2	\$350.00	
3	3	\$500.00	
4	1	\$500.00	
5	2	\$350.00	
6	5	\$250.00	
7	4	\$0.00	
8	0	\$0.00	

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Figure 1.13 The Project Equipment List as a Spreadsheet

ProjectName	OwnerContact	Phone	Category	Quantity	ItemDescription	UnitPrice	ExtendedPrice	Supplier	Discount
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	1	200 Amp Panel	\$170.00	\$170.00	NW Electric	25.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	1	300 Watt Breaker	\$60.00	\$180.00	NW Electric	25.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	7	720 Watt Breaker	\$55.00	\$245.00	NW Electric	25.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	15	15 Watt Breaker	\$55.00	\$25.00	NW Electric	25.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	200	12 ga. 3 Wire, per foot	\$1.50	\$300.00	EB Supplies	15.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	300	14 ga. 3 Wire, per foot	\$1.25	\$375.00	EB Supplies	15.00%
Baker Remodel	John Stanley	555-787-8392	Exterior	35	Siding, 4x8 feet	\$22.50	\$787.50	Contractor, Inc.	35.00%
Highland House	Elizabeth Barnaby	555-444-8899	Electrical	10	10 Watt Breaker	\$55.00	\$550.00	EB Supplies	15.00%
Baker Remodel	John Stanley	555-787-8392	Exterior	28	1x4 - 8 feet	\$4.75	\$133.00	Contractor, Inc.	35.00%
Baker Remodel	John Stanley	555-787-8392	Exterior	100	Cedar Shingles, bundle	\$65.00	\$6,500.00	Contractor, Inc.	35.00%
Highland House	Elizabeth Barnaby	555-444-8899	Interior	15	Door Hinge Set	\$29.95	\$449.25	Interior, Inc.	15.00%
Highland House	Elizabeth Barnaby	555-444-8899	Interior	15	Door Handle Set	\$52.50	\$787.50	Interior, Inc.	15.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	1	Panel, 200 Amp	\$170.00	\$170.00	NW Electric	25.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	2	50 Watt Breaker	\$60.00	\$120.00	NW Electric	25.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	5	20 Watt Breaker	\$35.00	\$175.00	NW Electric	25.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	20	15 Watt Breaker	\$35.00	\$700.00	NW Electric	25.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	150	12 ga. 3 Wire, per foot	\$1.50	\$225.00	NW Electric	25.00%
Hew Remodel	Ralph Hew	555-298-4244	Electrical	300	14 ga. 3 Wire, per foot	\$1.25	\$375.00	NW Electric	25.00%

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Figure 1.14 The Project Equipment Database Tables

ProjectID	ProjectName	OwnerContact	Phone
1	Highland House	Elizabeth Barnaby	555-444-8899
2	Baker Remodel	John Stanley	555-787-8392
3	Hew Remodel	Ralph Hew	555-298-4244

ItemNumber	ItemDescription	Category
100	200 Amp Panel	Electrical
1200	30 Watt Breaker	Electrical
1300	50 Watt Breaker	Electrical
1400	75 Watt Breaker	Electrical
1500	12 ga. 3 Wire, per foot	Electrical
1600	14 ga. 3 Wire, per foot	Electrical
1700	Siding, 4x8 feet	Exterior
1800	Cedar Shingles, bundle	Exterior
2000	Door	Interior
2100	Door Hinge Set	Interior
2200	Door Handle Set	Interior

SupplierID	Supplier	Discount
1	NW Electric	25.00%
2	EB Supplies	15.00%
3	Contractor, Inc.	35.00%
4	Interior, Inc.	15.00%

QuoteID	ProjectID	ItemNumber	SupplierID	Quantity	UnitPrice	ExtendedPrice
1	1	100	1	1	\$170.00	\$170.00
2	1	1200	1	7	\$60.00	\$420.00
3	1	1300	1	15	\$55.00	\$825.00
4	1	1400	1	15	\$55.00	\$825.00
5	1	1500	2	200	\$1.50	\$300.00
6	1	1600	2	300	\$1.25	\$375.00
7	2	1800	3	100	\$65.00	\$6,500.00
8	1	1400	2	10	\$55.00	\$550.00
9	2	1700	3	35	\$22.50	\$787.50
10	2	1800	3	100	\$65.00	\$6,500.00
11	1	2000	4	15	\$29.95	\$449.25
12	1	2100	4	15	\$52.50	\$787.50
13	1	2200	4	15	\$52.50	\$787.50
14	3	1000	1	1	\$170.00	\$170.00
15	3	1200	1	2	\$60.00	\$120.00
16	3	1300	1	5	\$55.00	\$275.00
17	3	1400	1	20	\$55.00	\$1,100.00
18	3	1500	1	150	\$1.50	\$225.00
19	3	1600	1	300	\$1.25	\$375.00

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How do I Process Relational Tables Using SQL?

- The leading technique for data definition and manipulation is **Structured Query Language (SQL)**.
- **SQL** is an international standard for creating, processing, and querying databases and their tables.
- Using **SQL** you can:
 - Reconstruct lists from their underlying tables,
 - Query for specific data conditions,
 - Perform calculations on data in tables, and
 - Insert, update, and delete data.



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SQL Art Course Database Example

Understand how using related tables helps you avoid the problems of using lists

- Using SQL, the following code will combine the three tables in the Art Course Database as seen in Figure 1.12:

```
SELECT    CUSTOMER.CustomerLastName,
          CUSTOMER.CustomerFirstName, CUSTOMER.Phone,
          COURSE.CourseDate, ENROLLMENT.AmountPaid, COURSE.Course,
          COURSE.Fee
FROM      CUSTOMER, ENROLLMENT, COURSE
WHERE     CUSTOMER.CustomerNumber = ENROLLMENT.CustomerNumber
          AND
          COURSE.CourseNumber = ENROLLMENT.CourseNumber;
```

- The results of running the code above can be seen in Figure 1.15 on the next page.



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Figure 1.15 Results of the SQL Query to Recreate the Art Course List Data

CustomerLastName	CustomerFirstName	Phone	CourseDate	AmountPaid	Course	Fee
Johnson	Ariel	206-567-1234	10/1/2019	\$250.00	Adv Pastels	\$500.00
Johnson	Ariel	206-567-1234	3/15/2019	\$350.00	Int Pastels	\$350.00
Green	Robin	425-678-8765	9/15/2019	\$350.00	Beg Oils	\$350.00
Jackson	Charles	360-789-3456	10/1/2019	\$500.00	Adv Pastels	\$500.00
Pearson	Jeffery	206-567-2345	10/1/2019	\$500.00	Adv Pastels	\$500.00
Sears	Miguel	360-789-4567	9/15/2019	\$350.00	Beg Oils	\$350.00
Kyle	Leah	425-678-7654	11/15/2019	\$250.00	Adv Pastels	\$500.00
Myers	Lynda	360-789-5678	10/15/2019	\$0.00	Beg Oils	\$350.00

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What is a Database System?

Know the components of a database system

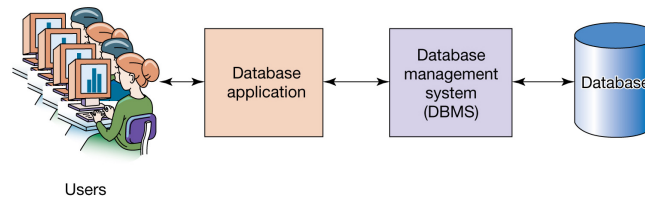
- A **database system** has four components consisting of:
 1. Users,
 2. Database application
 3. Database management system (DBMS), and
 4. Database.



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Figure 1.17 Components of a Database System

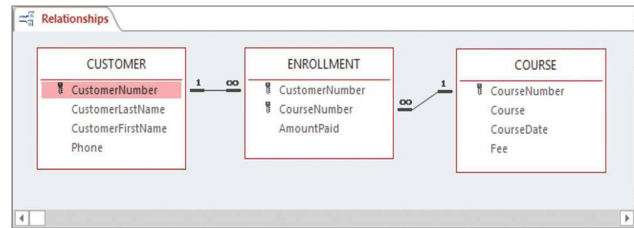


The Database

Know the components of a database system

- A **user** of a database system will:
 - Employ a database application to keep track of things
 - Use forms to read, enter, and query data
 - Produce reports
- A **database** is a self-describing collection of related tables:
 - **Self-describing** means a description of the structure of the database is contained with the database itself.
 - **Metadata** is data about the structure of the database.

Figure 1.18 Example Metadata: A Relationship Diagram for the Art Course Tables in Figure 1.12



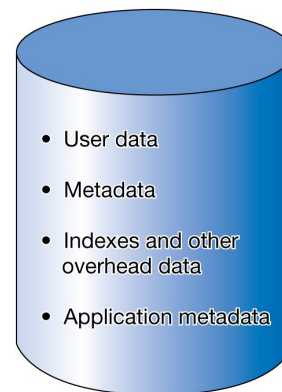
Access 2019, Windows 10, Microsoft Corporation.



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Figure 1.19 Database Contents



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Figure 1.20 Functions of a DBMS

Learn to purpose of a database management system (DBMS)

- The purpose of a DBMS is to create, process, and administer databases and are licensed from a vendor.
- The functions of a DBMS are shown in the following slide:

- Create database
- Create tables
- Create supporting structures (e.g., indexes)
- Read database data
- Modify (insert, update, or delete) database data
- Maintain database structures
- Enforce rules
- Control concurrency
- Provide security
- Perform backup and recovery

Referential Integrity Constraints

Learn the purpose of a database management system (DBMS)

- **Referential integrity constraints** are rules enforced by the DBMS to ensure values of a column in one table are valid when compared to values in another table.
 - For example, in the Art Course database what would happen if a user mistakenly entered 9 for CustomerNumber in the ENROLLMENT table?
 - Since 9 does not exist in that table it would cause errors and not execute.
 - To prevent this situation, the DBMS enforces the rule that if a CustomerNumber is entered in the ENROLLMENT table, must also exist in the CUSTOMER table.

Figure 1.21 Functions of Database Application Programs

- Create and process forms
- Process user queries
- Create and process reports
- Execute application logic
- Control application



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Figure 1.22 Example Data Entry Form

Course	CourseDate	Fee	AmountPaid	AmountDue
Int Pastels	3/15/2019	\$350.00	\$350.00	\$0.00
Adv Pastels	10/1/2019	\$500.00	\$250.00	\$250.00

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Figure 1.23 (a) Query Parameter Form, and (b) Query Results

Enter Parameter Value ? X

Enter part of course name:

pas

OK Cancel

Course Parameter Query

CustomerLastName	CustomerFirstName	Course	CourseDate	Fee	AmountPaid	Amount Due
Jackson	Charles	Adv Pastels	10/1/2019	\$500.00	\$500.00	\$0.00
Johnson	Ariel	Int Pastels	3/15/2019	\$350.00	\$350.00	\$0.00
Johnson	Ariel	Adv Pastels	10/1/2019	\$500.00	\$250.00	\$250.00
Kyle	Leah	Adv Pastels	11/15/2019	\$500.00	\$250.00	\$250.00
Pearson	Jeffery	Adv Pastels	10/1/2019	\$500.00	\$500.00	\$0.00

Record: 1 of 5 No Filter Search

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Figure 1.24 Example Report

Course Enrollment Report

Course	CourseDate	CustomerLastName	CustomerFirstName	Phone	Fee	AmountPaid	AmountDue
Adv Pastels							
	10/1/2019	Jackson	Charles	360-789-3456	\$500.00	\$500.00	\$0.00
		Johnson	Ariel	206-567-1234	\$500.00	\$250.00	\$250.00
		Pearson	Jeffery	206-567-2345	\$500.00	\$500.00	\$0.00
	11/15/2019	Kyle	Leah	425-678-7654	\$500.00	\$250.00	\$250.00
Reg Oils							
	9/15/2019	Green	Robin	425-678-8765	\$350.00	\$350.00	\$0.00
		Sears	Miguel	360-789-4567	\$350.00	\$350.00	\$0.00
	10/15/2019	Myers	Lynda	360-789-5678	\$350.00	\$0.00	\$350.00
Int Pastels							
	3/15/2019	Johnson	Ariel	206-567-1234	\$350.00	\$350.00	\$0.00

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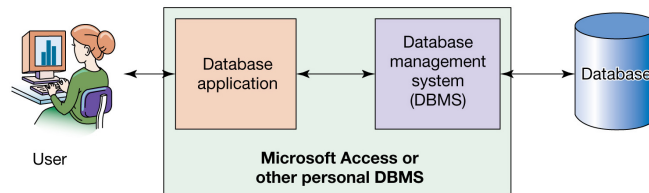
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Personal vs. Enterprise-Class Database Systems

- A **personal database system** is used by only one person and would include:
 - Only a few tables containing only a few hundred rows of data
 - Use only one computer with one user at a time
- An **enterprise database system is used international organizations with thousands** of concurrent users and would include:
 - Hundreds of tables with millions of rows of data
 - In use 24/7 24 hours a day

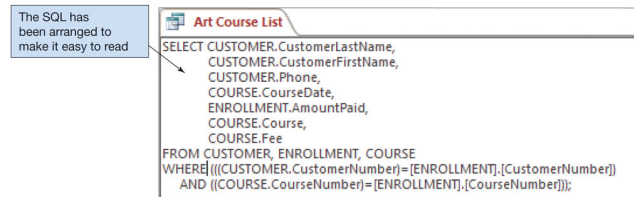
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Figure 1.25 Personal Database System



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Figure 1.26 SQL Generated by Microsoft Access Query



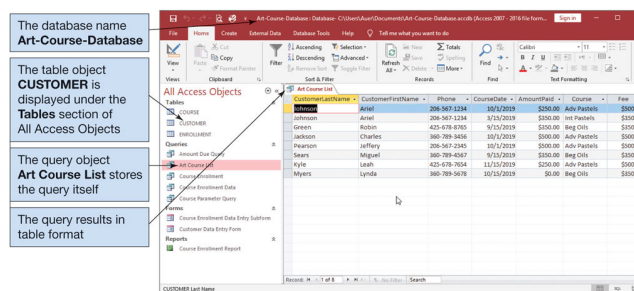
Access 2019, Windows 10, Microsoft Corporation.



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Figure 1.27 Microsoft Access 2019



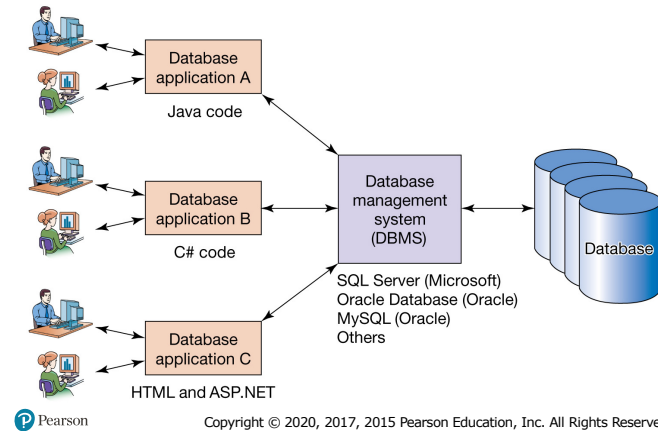
Access 2019, Windows 10, Microsoft Corporation.



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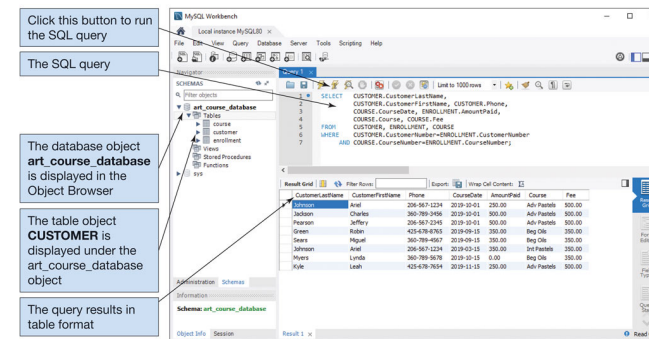
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Figure 1.28 Enterprise-Class Database System



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Figure 1.29 My SQL 8.0



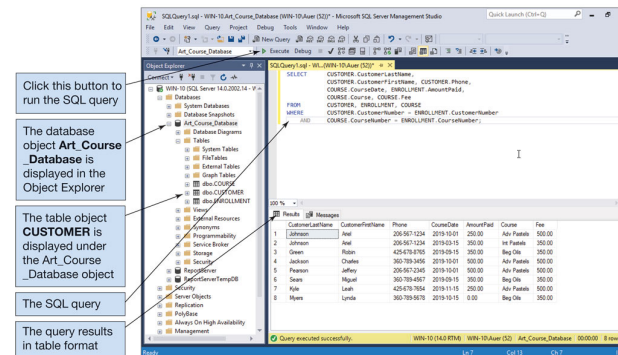
MySQL Community Server 8.0, MySQL Workbench, Oracle Corporation.

Pearson

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Figure 1.30 Microsoft SQL Server 2017



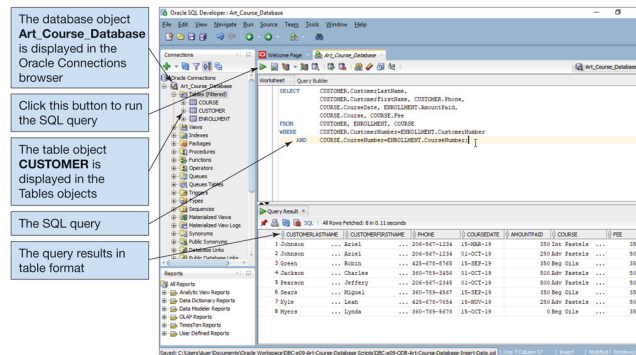
SQL Server 2017, SQL Server Management Studio, Microsoft Corporation.

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Figure 1.31 Oracle Database XE



Oracle Database XE, SQL Developer 18.4, Oracle Corporation.

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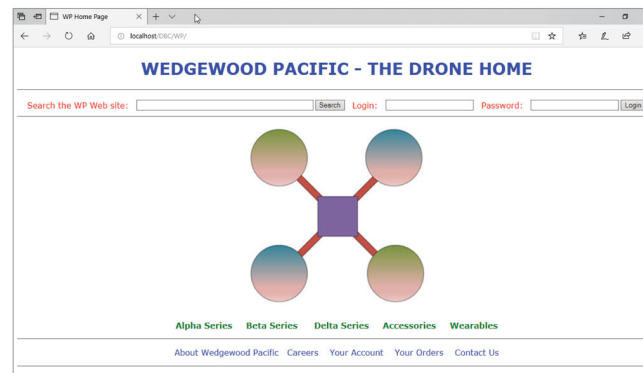
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What is a Web Database Application?

Introduce Web database applications

- A **Web database application** is an application with a Web user interface that is dependent on a database to store the data needed by the application (see Figure 1.32).
- An application programming interface (API) is a programming language such as PHP or JavaScript to connect to a DBMS allowing the sending of SQL commands to the DBMS and then to receive them back.

Figure 1.32 The WP Web Page



Microsoft Edge, Windows 10, Microsoft Corporation.

What are Data Warehouses and Business Intelligence (BI) Systems?

Introduce data warehouses and business intelligence (BI) systems

- **Transactions** are purchases bought online that are recorded in a company's database, also referred to as an **online transaction processing (OLTP)** database.
- Data analysis is done on an organization's **online analytical processing (OLAP)** database and is used for research.
- A **business intelligence system** consists of tools used to analyze and report on company data.



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What is Big Data?

Introduce Big Data and cloud computing

- **Big Data** is the current term for the enormous datasets generated by Web and mobile applications.
- **Nonrelational databases** are used to store Big Data (also known as NoSQL).



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Figure 1.33 ArangoDB

The query written in ArangoDB query language (AQL)

The query results in table format—note the extra columns _key, _id, and _rev

_key	_id	_rev	CourseNumber	Course	CourseDate	Fee	Events
17182940	COURSES/17182940	2	Reg	Reg	5/1/2019	300	[{"CustomerNumber": "C", "CustomerLastName": "Green", "CustomerFirstName": "Robert", "Phone": "432-678"}, {"CustomerNumber": "D", "CustomerLastName": "Brown", "CustomerFirstName": "Susan", "Phone": "789-456"}, {"CustomerNumber": "E", "CustomerLastName": "White", "CustomerFirstName": "John", "Phone": "123-456"}]
17182941	COURSES/17182941	4	Reg	Reg	10/1/2019	300	[{"CustomerNumber": "F", "CustomerLastName": "Brown", "CustomerFirstName": "Susan", "Phone": "789-456"}, {"CustomerNumber": "G", "CustomerLastName": "White", "CustomerFirstName": "John", "Phone": "123-456"}, {"CustomerNumber": "H", "CustomerLastName": "Black", "CustomerFirstName": "Jane", "Phone": "987-654"}]
17182942	COURSES/17182942	1	Adv	Points	10/1/2019	800	[{"CustomerNumber": "I", "CustomerLastName": "Brown", "CustomerFirstName": "Susan", "Phone": "789-456"}, {"CustomerNumber": "J", "CustomerLastName": "White", "CustomerFirstName": "John", "Phone": "123-456"}, {"CustomerNumber": "K", "CustomerLastName": "Black", "CustomerFirstName": "Jane", "Phone": "987-654"}]
17182943	COURSES/17182943	3	Adv	Points	5/1/2019	300	[{"CustomerNumber": "L", "CustomerLastName": "Brown", "CustomerFirstName": "Susan", "Phone": "789-456"}, {"CustomerNumber": "M", "CustomerLastName": "White", "CustomerFirstName": "John", "Phone": "123-456"}, {"CustomerNumber": "N", "CustomerLastName": "Black", "CustomerFirstName": "Jane", "Phone": "987-654"}]
17182944	COURSES/17182944	5	Adv	Points	11/1/2019	800	[{"CustomerNumber": "O", "CustomerLastName": "Brown", "CustomerFirstName": "Susan", "Phone": "789-456"}, {"CustomerNumber": "P", "CustomerLastName": "White", "CustomerFirstName": "John", "Phone": "123-456"}, {"CustomerNumber": "Q", "CustomerLastName": "Black", "CustomerFirstName": "Jane", "Phone": "987-654"}]

Courtesy of ArangoDB GmbH.



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What is Cloud Computing?

Introduce Big Data and cloud computing

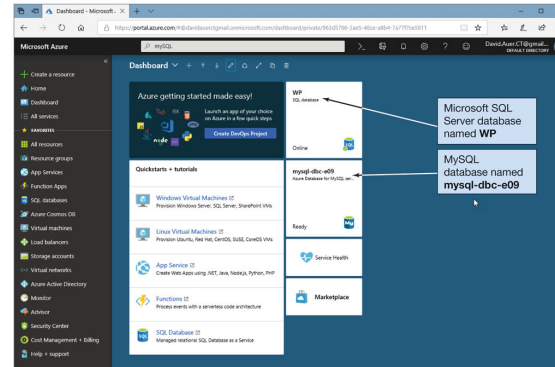
- **Cloud Computing** is the use of another company's hardware to conduct business via an Internet connection or through a Web server.
- Examples include:
 - Amazon Web Services (AWS)
 - Google Cloud Platform



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Figure 1.34 Microsoft Azure Cloud Service



Azure, Microsoft Edge, Microsoft Corporation.



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